

# MQL5 Expert Advisor Development Prompt

You are an expert MQL5 software engineer and quantitative trading systems developer. Your task is to build a **professional, institutional-quality, production-ready Expert Advisor (EA)** for MetaTrader 5.

The EA **must compile with ZERO errors and ZERO warnings** in the latest MetaEditor.

Do not leave placeholders, TODOs, pseudo-code, or incomplete functions. Every function must be fully implemented.

Before writing any code, thoroughly review the official MQL5 documentation and use best practices.

## Mandatory Requirement

**Make sure to go through MQL5 documentation and generate errorfree and correct code.**

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## General Requirements

- Use MQL5 only.
- Use `#property strict`.
- Use modern Object-Oriented Programming.
- Follow SOLID principles wherever practical.
- Produce clean, modular, maintainable code.
- Every class must have a single responsibility.
- Avoid duplicate code.
- Use constants and enums.
- Implement extensive comments.
- Never repaint.
- Never use future candles.
- Every trading decision must use ONLY CLOSED candles.
- Compatible with MT5 Strategy Tester.
- Fully optimization compatible.
- No global hidden state.
- Robust exception/error handling.
- Extensive logging.
- Validate every indicator handle.
- Validate copied historical data.
- Detect broker-supported filling mode automatically.
- Support IOC, FOK and RETURN filling modes.
- Handle 3, 4, 5 digit Forex brokers automatically.
- Handle Gold pricing correctly.
- Normalize prices using `SymbolInfoDouble`.
- Normalize lot size using broker limits.

- Never crash on missing data.
- Always fail gracefully.

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## Project Structure

Generate a complete modular project using the following structure.

```
TGCapitalEA.mq5

Include/

Constants.mqh
Enums.mqh
Utilities.mqh
Logger.mqh
BrokerInfo.mqh
DataCache.mqh
SymbolManager.mqh
SessionManager.mqh
EMAFilter.mqh
FVGDetector.mqh
DojiDetector.mqh
ConfirmationValidator.mqh
RiskManager.mqh
TradeManager.mqh
VisualizationManager.mqh
StateMachine.mqh
Version.mqh
```

Every file must be fully implemented.

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## Classes

Create fully implemented classes.

```
Logger

BrokerInfo

DataCache
```

SymbolManager  
SessionManager  
EMAFilter  
FVGDetector  
DojiDetector  
ConfirmationValidator  
RiskManager  
TradeManager  
VisualizationManager  
StateMachine

No empty classes.

No placeholders.

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## Strategy Name

TG Capital London EMA Stack + FVG Trident Strategy

Version 2.0

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## Symbols

Trade all enabled symbols independently.

Default symbols:

XAUUSD  
EURUSD  
USDJPY  
GBPUSD  
AUDUSD  
USDCAD

USDCHF  
EURGBP

Input must allow custom symbol list.

Automatically call SymbolSelect().

One-chart multi-symbol scanner.

Each symbol has its own independent state machine.

Maximum:

- One open trade per symbol.
- One pending order per symbol.
- One pending order per Fair Value Gap.

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## Multi-symbol Execution Rule

If multiple symbols generate valid entries on the SAME completed M30 candle:

Execute ONLY the FIRST valid symbol according to the order in the Symbols input list.

Ignore all other symbols for that M30 candle.

Those symbols may trade on later candles if a valid setup still exists.

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## Timeframes

Trend timeframe:

Daily

Execution timeframe:

M30

## Trading Session

Allow NEW trades ONLY during

03:00

until

06:30

America/New\_York

Requirements:

Automatically detect DST.

Convert New York time to broker server time.

Pending orders expire EXACTLY at

06:30:00 America/New\_York

even if the M30 candle is still open.

Existing trades remain open.

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## Trade Direction

Long only.

Never sell.

Never open short positions.

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## Daily Trend Filter

Only use COMPLETED Daily candle.

Conditions:

Close > EMA200

EMA5 > EMA9 > EMA13 > EMA21

EMA calculations:

Close price

If ANY condition fails:

Reject setup.

Cancel pending order immediately after the new Daily candle opens if filter becomes invalid.

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## Indicators

EMA5

EMA9

EMA13

EMA21

EMA200

All Close price.

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## Fair Value Gap

Bullish only.

Use THREE completed candles.

Definitions

A = oldest

B = middle

C = newest

Bullish FVG:

$\text{High}(A) < \text{Low}(C)$

Top

$\text{Low}(C)$

Bottom

$\text{High}(A)$

Size

$\text{Top} - \text{Bottom}$

Reject if

$\text{Size} \leq 0$

Consequent Encroachment

$\text{CE} = (\text{Top} + \text{Bottom}) / 2$

Only newest valid FVG remains active.

If newer FVG forms before entry:

Cancel pending order.

Replace with newest FVG.

Do NOT merge overlapping FVGs.

No overlap processing.

Assign every FVG a deterministic unique ID using:

- Symbol
- Candle C time
- FVG Top
- FVG Bottom

This ID prevents duplicate pending orders while allowing the same FVG to be traded again later if permitted.

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## Doji

### Definition

$$\begin{aligned} & \text{abs}(\text{Open} - \text{Close}) \\ & \leq \\ & \text{Threshold} \\ & \times \\ & (\text{High} - \text{Low}) \end{aligned}$$

Threshold input.

### Default

0.10

### Rules

Doji must occur AFTER FVG forms.

Low(Doji)

$\leq$  CE

Close(Doji)

CE

Otherwise reject.

If multiple Dojis occur:

Use FIRST valid Doji only.

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# Confirmation Candle

Must be the IMMEDIATELY NEXT completed M30 candle after the Doji.

Valid only if

Close(Confirmation)

<

High(Doji)

Otherwise reject.

---

## Entry

After confirmation candle closes

Place

BUY LIMIT

Entry

FVG Top

Never convert pending order to market order.

One Buy Limit per FVG.

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## Pending Order Expiry

Cancel immediately when either occurs first

1.

Exactly

06:30:00

America/New\_York

2.

Daily EMA filter fails on completed Daily candle.

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## Stop Loss

Low of Candle B

No buffer.

No ATR.

No spread adjustment.

Never modify stop loss.

Never trail stop.

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## Maximum Stop Distance

Inputs

Gold

600 points

Forex

100 pips

Reject trades exceeding limits.

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## Risk

Risk %

Input

Default

1%

Formula

```
Lot
=
(Account Balance × Risk%)
/
Risk Distance
```

Use Account Balance.

Normalize lots.

Respect

Minimum lot

Maximum lot

Lot step

Always FLOOR lot size so actual risk never exceeds requested risk.

Reject if normalized lot becomes zero.

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## Spread

Ignore spread completely.

No spread filter.

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## Slippage

Input

Default

5 points

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## Trade Management

No Take Profit.

No Break-even.

No Partial Close.

No Trailing Stop.

Hold trade until Daily EMA stack breaks.

Exit when completed Daily candle satisfies ANY

$\text{EMA5} \leq \text{EMA9}$

OR

$\text{EMA9} \leq \text{EMA13}$

OR

$\text{EMA13} \leq \text{EMA21}$

Exit immediately when new Daily candle opens.

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## Reject Trade If

Outside session

Daily filter invalid

No FVG

No Doji

Doji invalid

Confirmation invalid

Risk distance  $\leq 0$

Stop exceeds maximum

Duplicate pending order

Existing trade on symbol

Pending order exists

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## State Machine

Create enum.

```
WAIT_FOR_DAILY_FILTER  
  
WAIT_FOR_SESSION  
  
WAIT_FOR_FVG  
  
WAIT_FOR_DOJI  
  
WAIT_FOR_CONFIRMATION  
  
PLACE_PENDING_ORDER  
  
WAIT_FOR_FILL  
  
MANAGE_POSITION  
  
EXIT
```

Each symbol maintains independent state.

Persist state across terminal restarts by reconstructing from:

- Existing positions
- Pending orders
- Active FVG
- Market structure

Do not reset to initial state unless no recoverable state exists.

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## Visualization

Input

Enable Visualization

When enabled draw

FVG Rectangle

CE Line

Doji Marker

Confirmation Marker

Entry Line

Stop Line

Pending Order Status

Current State

Daily EMA values

Objects update automatically.

Delete all objects automatically once setup finishes or is invalidated.

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## Inputs

Symbols List

Risk %

Magic Number

Trade Comment

Maximum Slippage

Maximum Stop Gold

Maximum Stop Forex

Doji Threshold

Visualization

Log Level

EMA Periods

Session Start

Session End

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## Logging

Create log levels

NONE

ERROR

INFO

DEBUG

DEBUG logs must include

Indicator values

EMA values

FVG coordinates

FVG ID

CE

Lot calculation

Risk

Stop distance

Broker digits

Session conversion

DST status

Trade requests

Trade results

Order fills

Pending cancellation

State transitions

History loading

Indicator validation

Errors

---

## Historical Scan

Maximum initialization scan

100 completed M30 candles.

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## Broker Support

Auto detect

Digits

Tick size

Tick value



Contract size

Volume step

Minimum lot

Maximum lot

Freeze level

Stops level

Execution mode

Filling mode

Margin mode

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## Trade Execution

Use

CTrade

Detect proper filling mode automatically.

Retry temporary trade errors where appropriate.

Provide detailed error messages.

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## Data Validation

Before every calculation verify

Indicator handles

CopyBuffer success

CopyRates success

Enough historical bars

Valid prices

Valid volumes

Reject trading when any validation fails.

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## Engineering Quality

Avoid magic numbers.

Use constexpr where possible.

Use descriptive names.

Separate interface and implementation cleanly.

Minimize repeated calculations.

Cache indicator values where practical.

Optimize for Strategy Tester performance.

No memory leaks.

Destroy all indicator handles.

Release resources in OnDeinit().

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## OnInit()

Validate inputs.

Select symbols.

Initialize all managers.

Create indicator handles.

Load history.

Initialize state machines.

Recover existing positions and pending orders.

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## OnTick()

Process symbols sequentially.

One symbol at a time.

If a trade has already been opened during the current completed M30 candle, skip entries on the remaining symbols until the next M30 candle.

Update state machine.

Manage pending orders.

Manage positions.

Update visualization.

Write logs.

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## OnTradeTransaction()

Synchronize internal state.

Detect fills.

Detect cancellations.

Detect manual closures.

Recover correctly.

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## Acceptance Tests

The EA must pass all of these.

✓ Zero compile errors

- ✓ Zero warnings
- ✓ No repaint
- ✓ No future candles
- ✓ One-chart multi-symbol scanner
- ✓ Independent state machine
- ✓ Automatic SymbolSelect()
- ✓ Automatic DST detection
- ✓ Correct broker time conversion
- ✓ Correct Daily EMA filter
- ✓ Correct Bullish FVG
- ✓ Correct CE
- ✓ Correct Doji detection
- ✓ Immediate next confirmation candle
- ✓ Buy Limit exactly at FVG Top
- ✓ Stop Loss exactly at Candle B Low
- ✓ Pending expires exactly at 06:30 New York
- ✓ Pending cancelled if Daily filter fails
- ✓ Newer FVG replaces previous pending order
- ✓ One pending order per symbol
- ✓ One open trade per symbol
- ✓ Only first symbol trades when multiple signals occur on the same M30 candle
- ✓ Never move Stop Loss

- ✓ No Take Profit
  - ✓ Exit only on Daily EMA stack break
  - ✓ Restart recovery
  - ✓ Strategy Tester compatible
  - ✓ Optimization compatible
  - ✓ Fully deterministic behavior
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## Final Requirement

Return the COMPLETE project with all `.mq5` and `.mqh` files fully implemented.

Do **not** truncate the code.

Do **not** omit any file.

Do **not** summarize.

Generate every file in full.

If the response exceeds the output limit, continue automatically in subsequent responses until every file has been fully generated.

The final project must compile successfully in MetaEditor with **zero errors and zero warnings** without requiring manual fixes.

**Make sure to go through MQL5 documentation and generate errorfree and correct code.**